PROJECTS & ASSIGNMENTS PORTFOLIO

Brian Colquhoun, Hugh O'Donnell and Partners (BCHOD) have successfully worked on several assignments in Zambia and elsewhere in the Southern Africa region since 1948.

When the need arises, and complexity of the project dictates, we are able to combine the talents available in all our region offices. The company has handled several projects on a joint venture basis with several local and internationally renowned consulting engineering firms.

Such arrangements have greatly exposed the company to modern techniques of design and project management. Some of the experiences in the region are as illustrated below:

Lesotho: Maseru Wastewater Project in Lesotho. The project involved preparation of a feasibility study in addition to identification of remedial measures to overcome deficiencies in the existing sewerage scheme. The project also included formulation of extension measures for the sewerage scheme to include currently non sewered areas of the Greater City of Maseru. This involved investigative operations directed at establishing the commercial viability and sustainability of the proposed project. The project was funded by the European Union through the Ministry of Finance.

Zimbabwe: 10 ML/day Bulk Water Supply for the Municipality of Chinhoyi. Works included design and construction supervision of access roads and related storm water drainage, Pumping Mains and Pump Stations.

Democratic Republic of Congo: Senga Mines, D.R.C. The project involved investigations, detailed designs and tender documentation for three (3) dams of total capacity of 5000 mega-litres.

The efforts of Brian Colquhoun, Hugh O'Donnell and Partners have contributed to the success of several infrastructure projects in Zambia. These include:

TRANSPORTATION & ROADS

- Lusaka Decongestion Project: 2017
- Rehabilitation of the Kalulushi to Lufwanyama (M18) road.

- Upgrading to Bituminous standard and rehabilitation of selected urban roads in Lusaka (Lot 7 & 9) formula 1 road.
- Design and Construction Supervision of the Mazyopa Trash Trap Access road under the Millennium Challenge Account (MCA) project – 2016
- Design and Construction supervision of the ZARI Mount Makulu roads - 2016
- Road Condition surveys and Bridge Infrastructure Surveys of the entire Road and Bridge network in Mozambique – 2010 - 2012
- Consultancy Services for the Design and construction/upgrading of selected Chalala Roads, Mosi-o-Tunya Road Extension, part of Shantumbu Road and South of Chilenje South newly constructed Road Parallel to ZESCO Pylons based on Integrated Construction Unit (ICU) method of works.
- Consultancy Services for the construction supervision of Road T004 from the Airport Turnoff to Luangwa Bridge in Lusaka Province
- Supervision of the Periodic Maintenance of D020 Kapatu to Kasama (from Kandondo to Tamina) at the Junction to Luwingu and Kasama) in Northern Province (72.4Km)
- Accelerated Urban Roads Rehabilitation Programme 2007 - Solwezi
- Accelerated Urban Roads Rehabilitation Programme Rehabilitation of Urban Roads in Mongu
- Consultancy Services for Supervision of Periodic Maintenance works of T005 from Solwezi via Mutanda-Mwinilunga in North-Western Province
- Consulting Services for the Supervision of the Rehabilitation of Road T2 between Lusaka and Chirundu
- Accelerated Urban Roads Rehabilitation Programme in Livingstone
- Supervision of Periodic Maintenance of Stretches Between Km 0 and Km 36, Km 102 and Km 168 of the Road T5 between Chingola and Solwezi

Accelerated urban roads rehabilitation programme 2006/2007 – Mongu Town.

Mongu Accelerated Urban Roads Project was carried out on a Design and Build basis. There were six selected roads namely President Avenue, Airport Road, Libonda road, Arthur Wina Road, Kambule Street and Hospital Road, with total length of 8.13km and 2,100sm car park, of the 8.13 Km, 3 km was rehabilitation and 5.13Km was full construction. The Project value was US $1.4million

Arthur Wina Road before

The main works included the civil engineering design, supervision and construction of bituminous surface dressing, rehabilitation and reconstruction of existing pavement or gravel wearing course and replenishing or reshaping shoulders and roadside drainage.

Arthur Wina Road, Laying of the Second seal (Slurry)

Field investigation included Visual survey, traffic survey, topography survey and material survey of the existing pavement and prospective borrow pits.

In-situ testing and laboratory testing were carried out on sample of the pavement and the borrow pit.

On new construction, the works were carried by compacting existing base as subgrade, imported gravel for subbase and base. The Base was stabilized with 3% cement before compaction. The surface applied was Cape Seal.

Pavement Assessment using the FWD

Rehabilitation of Mukobeko Road and other township roads were carried out under the periodic maintenance of T2 Lusaka- Kabwe Road funded by European Union.

Mukobeko Road was in a very poor condition. It was badly potholed and had lost surface dressing on the last approximately 4.0Km section.

Assessment of the existing Pavement was carried using the Falling weight Deflectometer (FWD) and

the Dynamic cone Penetrometer (DCP). Manual traffic counts were carried out for seven days and an economic study was carried out to project the future expected traffic loading

Paver, Laying of 30mm thick Hot Premix Concrete Asphalt.

Mixing of 3% Cement to Gravel before compacting to a depth of 150mm thick as base.

The consultant produced Drawings, Bill of Quantities, Cost Estimates, and Technical Specifications.

The Interventions recommended were to scarifying the existing base, importing some material, mixing and compacting as subbase and then import additional material and lay a new 3% cement stabilised base.

Hot Premix Asphalt on the dual carriageway

The surface received 30mm thick Hot Premix Asphalt concrete.

The Project included rehabilitating the dual carriageway passing through the main town centre and the connected roads. The road markings were also done to define the movement order of the vehicles on different junctions.

Accelerated Urban Roads Rehabilitation Programme 2004/2005 – Lusaka City

Lusaka Accelerated Urban Roads Project was carried out on a Design and Build basis. There were fourteen (14no.) selected roads namely Kabelenga, Tuleteka, Longolongo, Sadzu, seventh street, Reedbuck, Middleway, Leopards Lane, car park, of the 8.13 Km, 3 km was rehabilitation and 5.13Km was full construction.

The Project value was US $1.4 million. The main works included the civil engineering design, supervision and construction of Bituminous surface dressing, rehabilitation and reconstruction of existing pavement or gravel wearing course and replenishing or reshaping shoulders and roadside drainage.

Field investigations included Visual survey, traffic survey, topography survey and material survey of construction material.

In-situ testing and laboratory testing were carried out on sample of the pavement and the borrow pit. On new construction, the works were carried by compacting existing base as subgrade, imported gravel for subbase and base. The Base was stabilized with 3% cement before compaction. The surface applied was Cape Seal

Accelerated Urban Roads Rehabilitation Programme 2006/2007 – Solwezi

Solwezi Accelerated Urban Roads Rehabilitation is being carried out on a Design and Build Projects. The Project includes reconstructiing of 5 roads, 7.7km total length. All the selected roads are unsurfaced and will be upgraded to surfaced road.

These being unsurfaced roads, field investigations included In-situ testing (DCP), Laboratory testing of the existing road/pavement material to assess the residual strength. Traffic counts were carried out to ascertain the expected load on the roads.

The recommended interventions after engineering analysis and design were to construct new subbase and base and apply cape seal as surface; other works include replenishing or reshaping shoulders and

Drainage works to be improved on Kyankwakwa road

Insitu Testing (DCP) of the existing pavement to determine the existing residual strength of the subgrade.

roadsid

e drainage.

Selected Road before rehabilitation

Accelerated Urban Roads Rehabilitation Programme 2003/2005 – Livingstone

Livingstone Accelerated Urban roads rehabilitation was completed in 2005 December. It was carried out on a design and build basis and the contact value was US $ 1.5million. In order to achieve the objectives of the consultancy, field investigations were carried out with regard to the following aspects:

- Topographical surveys of the roads.
- Evaluation of the existing pavement.
- Survey and analysis of traffic.
- Evaluation of roadside drainage.

These beiing unsurfaced roads, field investigations included In-situ testing (DCP), Laboratory testing of the existing road/pavement material to assess the residual strength. Traffic counts were carried out to ascertain the expected load on the roads.

The recommended interventions after engineering analysis and design were to patch up potholes and seal all cracks then reseal with a Cape seal.

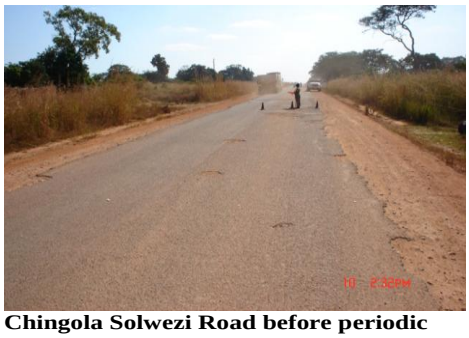
Only two roads required a reconstruction where the existing surface was scarified and compacted as subabse. A new 150mm thick stabilized base was laid and surfaced with a Cape seal. Constructed new subbase and base and apply cape seal as surface,

Other works include replenishing or reshaping shoulders and roadside drainage.

Periodic Maintenance of 102.6 Km of Road (T5) Chingola – Solwezi

Periodic Maintenance of Chingola Solwezi road was funded through the Government of Zambia, Ministry of Works and Supply, Road Fund Agency. The Implementing Agency was the Roads Development Agency.

The length of the road that was worked on was 102.6 km, starting from Chingola on the Copperbelt to Solwezi in North Western Province. Most of the road was in a very bad state, with a lot of pot-holes.



Chingola Solwezi Road before periodic maintenance

Several sections along the road required complete reconstruction; there were a lot of omissions in the original Bill of Quantities (BOQ) that led to variations,

for example, the tonnage of cement, cleaning and de-silting of culverts. Reshaping of side drains to invert

was only billed for the first stretch (km 0 + 000 to Km

36 + 600), there were no quantities for similar works from km 102 + 000 to Km 168 + 000.

The contract included earthworks, drainage works, shoulder repair, road furniture and markings and carriageway reconstruction, including bituminous surface treatments and dressings, broken down as follows:

Earthworks - Reshaping of the shoulder and side drain

Pot hole Patching prior to resurfcaing

Rehabilitation of selected core district and feeder roads in Mumbwa, Kaoma and Mongu Districts: 2005 - 2007

Rehabilitation of District and Feeder Roads in three districts namely Mumbwa, Kaoma and Mongu Districts was funded by the Danish International Development Agency (DANIDA) through Road Development Agency (RDA).

The Project commenced in 2005 and was scheduled to be completed by end of 2007. The Consultancy involved field surveys to identify improvement works required such as earthworks, gravelling requirements and drainage structures.It also included preparation of a confidetial Engineers estimate, Tender Documentation and works supervision.

The Project covered eight 9*No) roads giving a toatal road network of 245.oKm in the three districts.It involved both machine based technology (MBT) and Labour Based Technology (LBT)

The Supervision component involved guiding the contractors in setting out, earthworks, construction of drainage structures, gravelling and quality control.

Accelerated urban roads Rehabilitation programme 2006/2007 – Mansa Town.

Eroded Shoulders on Mwaiseni road

Mansa Accelerated Urban Roads Project was carried out on a Design and Build basis. The Project value was US $1.5million. The Project is expected to finish in June 2006.

The main works included the civil engineering design, supervision and construction of

Bituminous surface dressing, rehabilitation and reconstruction of existing pavement or gravel wearing course and replenishing or reshaping shoulders and roadside drainage.

Field investigation included Visual survey, traffic survey, topography survey and material survey of the existing pavement and prospective borrow pits.

In-situ testing and laboratory testing were carried out on sample of the pavement and the borrow pit.

On new construction, the works were carried by compacting existing base as subgrade, imported gravel for subbase and base. The Base was stabilized with 3% cement before compaction. The surface applied was Cape Seal.

Rehabilitation of Forest Feeder Roads in Luapula and North-Western Province

Rehabilitation of feeder roads in Luapula and North-Western Province was funded by International Fund for Agriculture Development (IFAD)as a sub-component of the Forest Resources Management Project (FRMP). The Project commenced in 2005 and is expected to finish end of 2007.

The works undertaken by the consultant were to assess the roads and design rehabilitation interventions, Cost estimate and drawing preparation, tender document preparation and evaluation the bids.

The Project covered six and five districts in Luapula and North-Western Province respectively. The Project was labour based. The main works carried out were setting out, site clearance, quarrying, earthworks, drainage works, reshaping of the roads to feeder roads standard, gravelling works and other miscellaneous works.

WATER & SANITATION

The firm has been, and continues to be involved in the implementation of several water and sanitation projects of different magnitudes and complexity. Some of the projects that have been successfully implemented include the following:

- Design and supervision of a water and sanitation improvement project in Muchinga Province under the Small Towns Integrated Water Supply and Sanitation Project funded by the AfDB.
- World Bank funded project involving delivery of 10,000 On-site Sanitation Systems (OSS) to two priority project areas of Kanyama and Chawama/Kuomboka townships in Lusaka under the Lusaka Sanitation Program (LSP).
- JICA funded project in Lusaka. The objective of the assignment was to find out the status quo of water resource management and public hygiene in the compound of Misisi.
- EIB funded project for Mulonga Water and Sewerage Company (MWSC). The project involved the provision of technical assistance to the water utility company and the key objective of the project (Zambia Water and Sanitation Project) was to rehabilitate and expand water and waste water services of Mulonga Water and Sewage Company (MWSC) and improve business capacity of the water company.
- Rehabilitation and extension of water and sanitation infrastructure in Nchelenge and Chiengi districts in Luapula Province. The project involved assessment of existing water and sanitation facilities to identify performance deficiencies within the systems found at the two towns. The assessment exercise was followed by design of corrective measures including expansion of the schemes to all unserved or new areas.
- Design and construction supervision of rehabilitation of water supply and sanitation infrastructure in Chisali District of Muchinga Province. The project included construction of a new dam and expansion of the water treatment plant and pump station.
- Rehabilitation Measures for Sewerage Facilities in Kansuswa Peri-Urban Area of Mufulira.
- Design and Supervision of Rehabilitation and Extension of water supply systems in the Periurban and Low-cost housing areas of Western Province in the towns of Mongu, Kaoma, Senanga, Sesheke, Kalabo, Lukulu, Limulunga and Namushakende.
- Immediate Restoration of Water and Sanitation Goods and Works for Lusaka Water and Sewerage Company (funded by the World Bank)
- Review of the Water and Sewerage Service Improvement Study for Nkana Water and Sewerage Company (NWSC).

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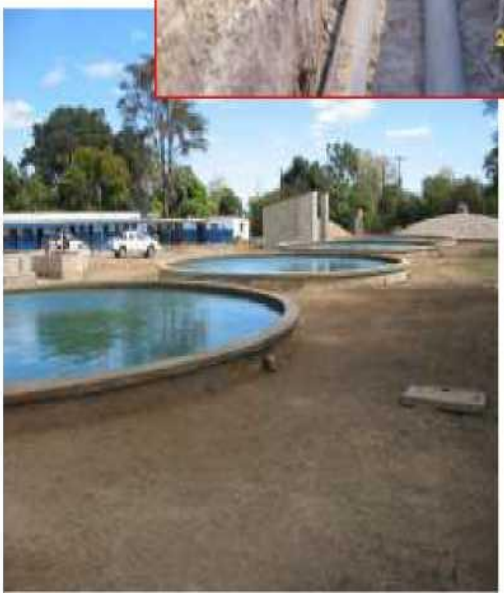

Irrigation Infrastructure

Feasibility Study, Design and Construction of the ***Mpongwe Irrigation Scheme*** on the Copperbelt Province

Engineering Design and Construction Supervision of Small-Scale Irrigation Proj ect (SIP) in Sinazongwe and Chongwe Districts. Scope of works included topographical surveys, design service tanks, open irrigation canals and drains networks, canal control structures, roads network within the schemes, main and booster pumping stations and bills of quantities, tender documents.

Hydrology

Flood calculations for the proposed Bridge across Mvuvye River in Nyimba District of Eastern Province of Zambia. The assignment involved field surveys of the catchment area, analysing the Mean Annual Precipitation (MAP) of the area, determining flood levels, detailed design of the Bridge and Tender Documentation

BUILDINGS & INFRASTRUCTURE DEVELOPMENT

We have extensive experience in building works in Zambia and in design procedures and contract administration required by private developers and government departments.

We have undertaken the following projects, in Zambia, in recent years:

- Office Block for Zambia Public Procurement Authority in Lusaka
- Cancer Diseases Hospital Phase II
- Levy Business Park
- Factory for HITACH along Airport Road in Lusaka
- Danish Embassy Car Park
- Chirundu Border Post
- IZWE building

LEVY
SAMSUNG
SAMSUNG

Project Name	Client Name	Type of Services	Funding Agency
Consultancy Services for the Techno-Economic Study, Detailed Engineering Design and Construction Supervision of the Water Supply and Sanitation Improvement Project in the Urban Centres of Muchinga Province	Ministry of Water Development and Sanitation (GRZ)	Design of water sources, transmission mains, storage facilities and distribution systems.	GRZ/ AfDB
Lusaka Sanitation Project (LSP) - Detailed Design and Construction, Supervision and Verification of Household Sanitation Facilities in Lusaka	Lusaka Water Supply and Sanitation Company (LWSC)	Project promotion and sensitization campaigns, undertaking designs of on-site sanitation facilities, marketing of sanitation facilities, project management and supervision of construction works.	The World Bank
The Investigation of a Compound in Lusaka for Water Resource and Public Hygiene	JICA	Collection of various forms of data and situation analysis in relation to water resources, drainage systems, and public hygiene.	JICA
Technical Assistance to the Zambia Water and Sanitation Project – Mulonga Water and Sewerage Company (MWSC)	Mulonga Water Supply and Sanitation Company (MWSC)	Conducting feasibility study, design operations, preparation of peri-urban plan, undertaking topographical surveys, conducting socio-economic surveys, etc.	European Investment Bank (EIB)
Design and Supervise the Expansion and Rehabilitation of Water and Sewerage Infrastructure in Nchelenge and Chiengi	The Ministry of Local Government and Housing	Undertook design and supervision operations.	GRZ

Central Province Water and Sanitation Project	Ministry of Local Government and Housing	Carried out field surveys and were responsible for preparation of both preliminary and final designs including production of working drawings. Also prepared investment costs including contract documents in preparation for tendering. This was a massive project that required	African Development Bank (AfDB)
Study of Water and Sewerage Service Improvement in Kitwe City, Kalulushi and Chambishi Towns	Nkana Water and Sewerage Company	Assessed the extent and cost implication of needed rehabilitation works in the three towns by physical site inspections and review of reports	African Development Bank (AfDB) and World Bank (WB)
Eight Centers Water Supply and Sanitation Project	Nkana Water and Sewerage Company	Health Care, Public Health and hygiene promotion/sensitization Communication and Community Mobilisation. Establishing demonstration latrines. ToT training in CLTS. HIV/AIDS awareness Education	African Development Bank (AfDB)
Detailed Engineering Design and Construction Supervision of the Water Supply and Sanitation Improvement Provement Project in Urban Centres of Muchinga Province (Mpika, Chinsali, Isoka and Nakonde)	Chambeshi Water and Sewerage Company	Feasibility studies, Topographic Surveys, Conditional Survey and Inventory of water and sanitation infrastructure, Designs and Tender Documentation	African Development Bank (AfDB)
Undertook siting, drilling and borehole installation in Lusaka, Southern and Western provinces (lot 1&2), over 400 no. boreholes.	The Ministry of Local Government and Housing (MLGH)	Conducted geophysical investigations and supervised the drilling operations of over 400 boreholes in the three provinces.	DANIDA

Project: Implementation of the water and sanitation project in southern province and the City of Livingstone.	The Ministry of Local Government and Housing (MLGH)	Undertook design operations and supervision of construction works.	KfW